



Lost opportunities: Coral recruitment does not translate to reef recovery in the Florida Keys



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ABSTRACT

We tested the hypothesis that the poor recovery of the coral populations on reefs in the Florida Keys is related to low coral recruitment. In the summer of 2011, we deployed 240 terracotta tiles at eight study sites in a balanced design: (i) among three depths; and (ii) between fished and unfished reefs. Corals recruited to ~40% of the deployed tiles, with more corals settling on tiles on unfished reefs than on fished reefs. The apparent effect of protection was not a consequence of different densities of herbivorous fishes, but was more likely related to local hydrography and the tendency of the no-take reserves to act as larval sinks, particularly in the lower Florida Keys. There was a mismatch between the coral taxa that recruited and the adult coral assemblages, suggesting that recruits were arriving but not surviving to contribute to coral recovery.

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1. Introduction

If coral populations are to persist, coral recruitment must equal or exceed the rates of mortality of adult corals. The generation of recruits, particularly those with novel genotypes derived through sexual recombination, is necessary to maintain and restore coral populations, especially when environments are changing rapidly (Richmond, 1997; Gardner et al., 2003; Ritson-Williams et al., 2010). The increasing frequency and severity of perturbations to coral reefs are, however, reducing coral populations worldwide (Bruno and Selig, 2007). As coral cover continues to decline throughout the Caribbean, the supply of coral planulae may be insufficient to maintain and replenish coral-dominated communities (Vermeij, 2006; Gleason and Hofmann, 2011).

In response to signs of fish-population declines throughout the Florida Keys, the 9500-km² Florida Keys National Marine Sanctuary (FKNMS) was established in 1990 (Florida Keys National Marine Sanctuary, 1996). In 1997, 24 no-take reserves, distributed from Miami to the Tortugas Banks, were established within the Sanctuary to protect fish stocks. Although the Florida Keys National Marine Sanctuary was set up, in part, to ensure the sustainable use of the Florida Keys by “achieving a balance between comprehensive resource protection and multiple, compatible uses of those resources” (Florida Keys National Marine Sanctuary, 1996, p. 5), the placement of the no-take reserves was not independent of

oceanographic setting (B. Causey, pers. comm.). There was already, in 1990, an understanding of the local hydrography. The reefs of the Dry Tortugas and the Florida Current interact to form a series of gyres. The largest and most frequently forming is the Tortugas Gyre, which extends up to 100 km in diameter and often persists for as long as 140 days. The smaller Pourtales Gyre is adjacent to Western Sambo Reef (Lee et al., 1992). The gyres diminish in size as they propagate to the northeast (Lee et al., 1992). Modeling studies have predicted that these gyres retain locally produced larvae that would otherwise be advected by the Florida Current (Lee et al., 1994; Criales and Lee, 1995). At least implicitly, there was potential for the no-take reserves in the FKNMS to concentrate larvae, hydrographically, and act as larval sinks.

A reasonable goal of marine reserves on coral reefs is that closing selected areas to fishing pressure should increase the biomass of target fish species. A less-tenable expectation is that preventing fishing pressure will increase stocks of herbivorous fishes, which will crop macroalgae and, therefore, improve the chances of coral recruitment and coral population recovery. Although the densities of carnivorous fishes have increased in the reserves (Kramer and Heck, 2007), there is no evidence that the reserves have had a beneficial, cascading effect on corals (Aronson and Precht, 2006; Schutte et al., 2010; Toth et al., 2014).

The recent decline of coral populations in the Florida Keys is thought to be largely a consequence of low coral recruitment (Tougas and Porter, 2002) or of mortality outpacing recruitment (Williams and Miller, 2012). To test these conjectures, we examined the spatial variation in coral recruitment in the Florida Keys

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in 2011. We deployed 240 terracotta tiles to determine whether corals were recruiting in large numbers or whether coral recruitment was indeed limiting the recovery of coral populations. We were also interested in whether coral recruitment differed between fished and unfished reefs, and whether shallow habitats received more recruits than deep habitats. We asked three overarching questions: (i) Does coral recruitment vary between levels of protection in the Florida Keys? (ii) Does coral recruitment vary among depths along the Florida reef tract? and (iii) Is there a difference in recruitment between the upper and lower Florida Keys?

2. Methods

2.1. Field methods

We examined patterns of coral recruitment at eight study sites along the Florida reef tract (Fig. 1). Four sites were established on unfished reefs and four on fished reefs. At each site, we installed 30 new, unglazed terracotta tiles ($10 \times 10 \times 1.5$ cm), deploying 10 tiles at each of three depths (2–5 m, 7–10 m, and 14–17 m). A total of 240 tiles were installed during the period 9–17 May 2011, 232 of which were retrieved during 23–27 September 2011 following a 133- to 141-day soaking period; these dates encompass the coral-spawning period for the Florida Keys (van Woesik et al., 2006).

Prior to attaching the tiles to the reef substrate, we drilled holes 0.5 cm in diameter into the non-living carbonate framework at randomly selected points using a pneumatic drill attached to a SCUBA tank. A ribbed, plastic, drywall anchor was inserted into each drilled hole, and a single stainless-steel screw (35 mm long $\times$ 3.2 mm wide) and stainless-steel washer (15 mm diameter and 1 mm thick) were used to attach the tile to the reef. Tiles were installed to match the reef contours to the greatest extent possible. Prior to removal, the angle of each tile was measured relative to the

horizontal, to determine if substrate angle influenced recruitment. The substrate angle varied from 0° to 85° , with an average angle of 18° (standard deviation 19°). The proportion of the undersurface of the tiles exposed to herbivorous fish depended on the substrate angle and reef undulations. The tiles were photographed underwater using a Canon Powershot SD790 IS digital camera. The tiles were removed from the reef, placed in individually marked plastic containers, and transported to the Florida Institute of Technology for analysis.

2.2. Laboratory methods

Coral recruits were identified to the lowest taxonomic level possible and photographed using an Olympus SZX12 Stereoscope and Nikon Coolpix 8700 digital camera. We were able to categorize the coral recruits into three genera, *Siderastrea*, *Acropora*, *Porites*; and two families, Astrocoeniidae and Meandrinidae (Fig. 2). We also categorized the orientation of each recruit on the tile (i.e., top, bottom, or side), and determined the proximity of each coral recruit to crustose coralline algae. To determine the percentage cover of sessile, benthic organisms on the bottom surfaces of the tiles, where the majority of recruits settled, photographs were taken using a Nikon D80 digital SLR camera with an 18–55 mm f/3.5–5.6 G AF-S DX VR Zoom-Nikkor lens. The images were processed using the software Coral-Point-Count with Excel extensions (CPCe) (Kohler and Gill, 2006). A pilot study showed that 20 random points per tile was adequate to make accurate assessments of coverage. We classified each point as one of the following: (i) bare substrate; (ii) suspension-feeders (ascidians, sponges, barnacles, serpulid worms, and bryozoans); (iii) macroalgae (>1 cm high); (iv) turf algae (≤ 1 cm high); or (v) crustose coralline algae. Recruits on the sides of the tiles were pooled with the recruits on the bottoms of the tiles. Only three recruits were observed on the tops of tiles, and they were not included in the analysis.

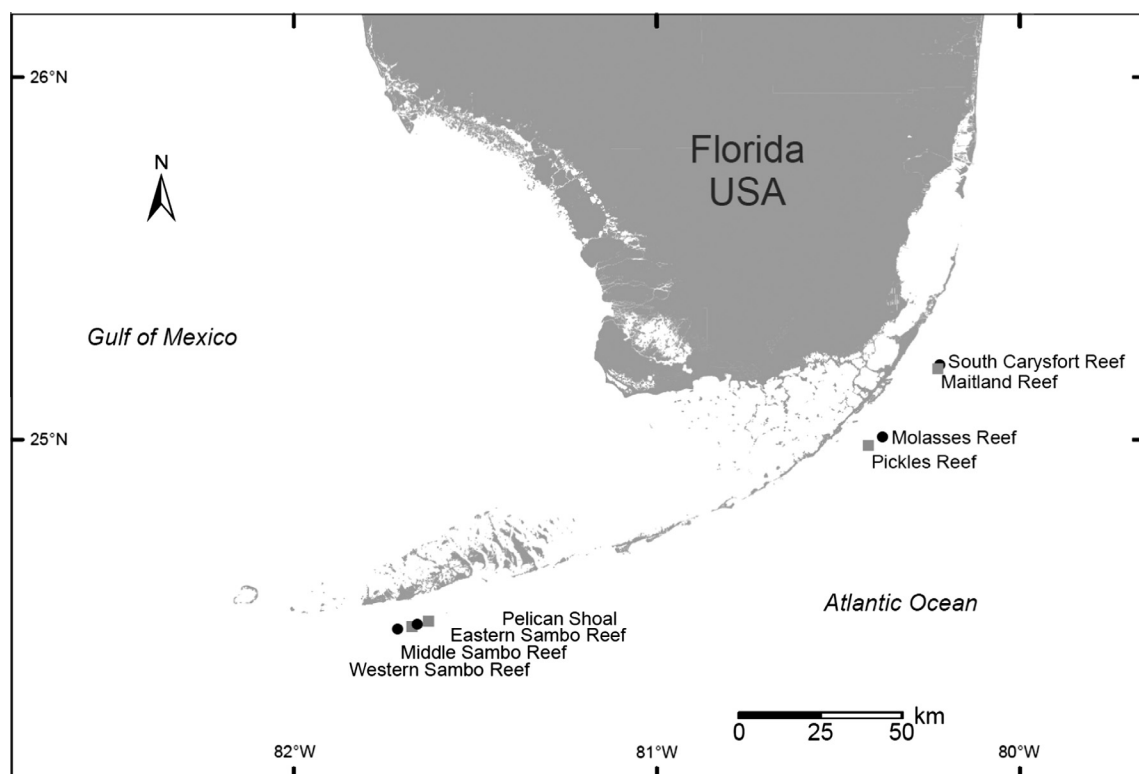


Fig. 1. Location of the eight study sites in the Florida Keys. Fished reefs are represented by gray squares and unfished reefs are represented by black circles.



Fig. 2. Representative images of coral recruits that settled on recruitment plates deployed in the Florida Keys in 2011. (a) *Siderastrea*: symmetrically-spaced septa, saw-like septal teeth, poorly defined corallite walls, and large columella; (b) *Acropora*: no columella and porous skeleton; (c) *Porites*: paliform lobes present, porous skeleton, and inconspicuous coenosteum; (d) *Astrocoeniidae*: well defined corallite walls, neatly arranged septa, columella present; (e) *Meandrinidae*: non-porous skeleton, and evenly spaced, solid septa.

2.3. Data analysis

We used a generalized linear model with a Poisson probability distribution to assess differential recruitment in the Florida Keys using protection, site, depth, and region as factors. The initial model included all main effects and all possible interactions. A back-step iteration process sequentially eliminated under-performing factors that did not improve the model, which included the elimination of site as a random variable. The final model was selected based on the lowest Akaike's Information Criterion, which only included significant factors. All factors were considered fixed, with the first factor, protection, defined at two levels as either fished or unfished; the second factor, depth, defined at three levels, 2–5 m, 7–10 m, and 14–17 m; and the third factor, region, defined at two levels, upper and lower Florida Keys. Settlement tiles acted as independent replicates, and the recruitment data were counts.

The relationships between major sessile, benthic groups and coral recruitment were assessed using a series of Poisson-distributed, generalized linear-regression models. All statistical procedures were completed using R ([R Core development team, 2012](#)) on a Windows platform.

3. Results

Contrary to the assumption of limited recruitment in the Florida Keys, we recorded coral recruits on nearly 40% of the tiles that were deployed. A total of 92 scleractinian coral recruits were found on the 232 terracotta tiles that were recovered from the Florida Keys in 2011. Coral recruits were mainly *Siderastrea* (26%), *Acropora* (26%), *Porites* (24%), *Astrocoeniidae* (21%), and *Meandrinidae* (3%). We did not find any *Orbicella* (formerly *Montastraea*) recruits on the tiles. There were significant differences between the relative

percentages of the coral recruits and the relative percentages of the adult coral assemblages reported by Toth et al. (2014) ($\chi^2 = 131.8$, $df = 5$, $p < 0.001$), mostly because the adult assemblages were missing *Acropora* and the recruit assemblages were missing *Orbicella*. The angles of the tiles on the reefs had no significant effect on coral recruitment ($\chi^2 = 2.93$, $df = 1$, $p = 0.087$).

Whether the tiles were placed in the upper or lower Florida Keys had no overall significant effect on coral recruitment (Fig. 3). There was, however, significantly higher coral recruitment at 2–5 m in the lower than in the upper Florida Keys ($p < 0.01$). There were no differences at other depths (Fig. 3). There were significantly more coral recruits on tiles placed in unfished areas than in fished areas of the Florida Keys ($p < 0.05$), but only between 2 and 10 m (Fig. 4a). Recruitment was highest at 7–10 m in unfished areas (Fig. 4a). There were no significant differences in macroalgal cover on tiles between fished and unfished reefs ($p > 0.05$; Fig. 4b). There were also no significant differences in coralline algae cover on tiles between fished and unfished reefs ($p > 0.05$; Fig. 4c). We noted that coral recruits preferentially settled in close proximity to (i.e., ≤ 10 mm from) crustose coralline algae (69%), rather than directly on the surfaces of crustose coralline algae (31%). The percentage cover of suspension-feeders (serpulid worms, sponges, barnacles, bryozoans, and ascidians) pooled after no significant relationships were found among these groups) was negatively related to coral recruitment at 2–5 m (slope = -0.023 , $p = 0.04$). There was no significant relationship between the density of coral recruits and the percentage cover of crustose coralline algae ($p > 0.05$). There was also no significant relationship between the density of coral recruits and the percentage cover of macroalgae ($p > 0.05$).

4. Discussion

Compared with historical accounts (Agassiz, 1880; Precht and Miller, 2007), coral cover has been low in the Florida Keys over the last few decades, averaging less than 10% (Murdoch and Aronson, 1999; Smith et al., 2009; Wagner et al., 2010; Schutte et al., 2010; Office of National Marine Sanctuaries, 2011). Such

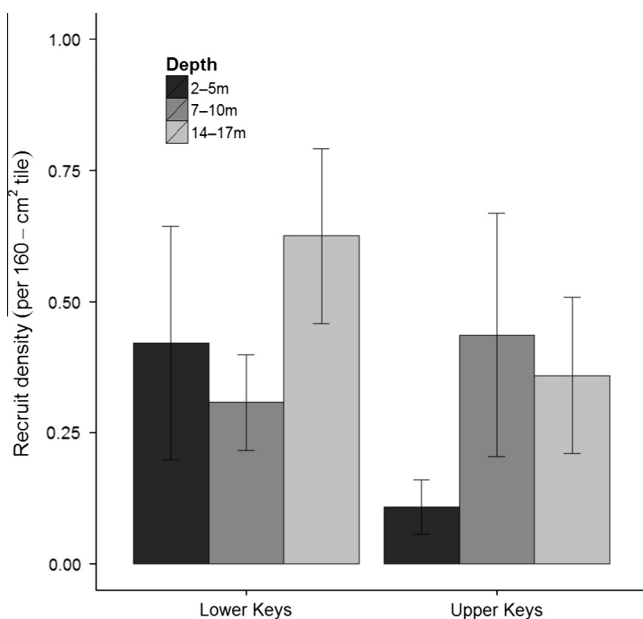


Fig. 3. Effect of region. Interaction plot displaying mean coral recruitment per settlement tile (± 1 standard error) and the effects of region (upper or lower Florida Keys) and depth (2–5 m, 7–10 m, and 14–17 m) on coral recruitment in the Florida Keys.

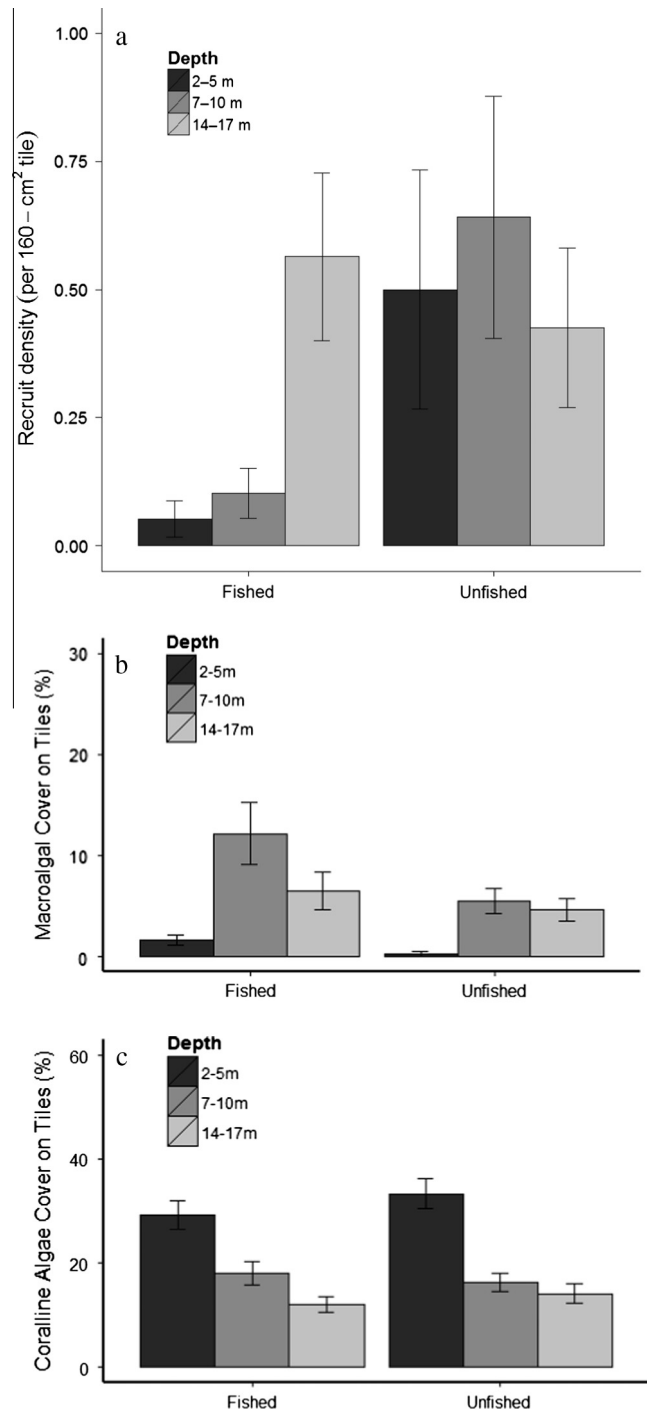


Fig. 4. Effects of protection and depth. (a) Interaction plot displaying mean coral recruitment (± 1 standard error) per settlement tile and the effects of protection (fished and unfished) and depth (2–5 m, 7–10 m, and 14–17 m) on coral recruitment in the Florida Keys. (b) Percentage cover of macroalgae (mean ± 1 standard error) on the bottom- and side-surfaces of settlement tiles (160 cm²), categorized by protection (fished and unfished) and depth (2–5 m, 7–10 m, and 14–17 m). (c) Percentage cover of coralline algae (mean ± 1 standard error) on the bottom- and side-surfaces of settlement tiles (160 cm²), categorized by protection (fished and unfished) and depth (2–5 m, 7–10 m, and 14–17 m).

low coverage of corals has led to the presumption that the reefs are not recovering because they have received few coral recruits (Tougas and Porter, 2002). That presumption may not be correct, considering that we observed on average 0.38 recruits per tile in the Florida Keys in 2011, many of which were *Acropora* recruits, a threatened genus in the Caribbean (Hogarth, 2006). Other studies

Table 1

Comparative summary of coral recruitment data for 22 studies on reefs in the Caribbean since 1979.

Location	Density of coral recruits	Corals	Definition	Methods	Depth (m)	Duration	Reference
Barbados	8–34 recruits/m ²	<i>Agaricia</i> , <i>Porites</i> , <i>Siderastrea siderea</i> , <i>Siderastrea radians</i> , <i>Stephanocoenia intersepta</i> , <i>Leptoseris cucullata</i> , <i>Madracis</i> , <i>Dichocoenia stokesi</i> , <i>Favia fragum</i> , <i>Manicina areolata</i> , <i>Diploria</i> , <i>Montastraea</i>	≤40 mm	1-m ² quadrats (<i>n</i> = 20) along a 40 m transect adjacent to transect used for analysis of community structure and <i>Diadema</i> populations in algal and urchin zones	<6	Single	Carpenter and Edmunds (2006)
Belize	4–8 recruits/m ²	<i>Agaricia</i> , <i>Porites</i> , <i>Siderastrea siderea</i> , <i>Siderastrea radians</i> , <i>Stephanocoenia intersepta</i> , <i>Leptoseris cucullata</i> , <i>Madracis</i> , <i>Dichocoenia stokesi</i> , <i>Favia fragum</i> , <i>Manicina areolata</i> , <i>Diploria</i> , <i>Montastraea</i>	Newly settled	1-m ² quadrats (<i>n</i> = 20) along a 40 m transect adjacent to transect used for analysis of community structure and <i>Diadema</i> populations in algal and urchin zones	10	12 months	Arnold and Steneck (2011)
Bonaire, Netherlands Antilles	0.3–2.26 spat per plate (yr ⁻¹)	<i>Agaricia</i> , <i>Porites</i>	≤40 mm	25 tiles at each site (<i>n</i> = 4)	5	Single	Bak and Engel (1979)
	19.4 recruits/m ²	<i>Agaricia agaricites</i> , <i>Leptoseris cucullata</i> , <i>Porites astreoides</i> , <i>Meandrina meandrites</i> , <i>Agaricia lamarcki</i>	≤40 mm	1-m ² quadrats (<i>n</i> = 5/depth)	11		
	20.2 recruits/m ²				29		
	13.2 recruits/m ²				<6		Carpenter and Edmunds (2006)
	7–11 recruits (m ⁻²)	<i>Agaricia</i> , <i>Porites</i> , <i>Siderastrea siderea</i> , <i>Siderastrea radians</i> , <i>Stephanocoenia intersepta</i> , <i>Leptoseris cucullata</i> , <i>Madracis</i> , <i>Dichocoenia stokesi</i> , <i>Favia fragum</i> , <i>Manicina areolata</i> , <i>Diploria</i> , <i>Montastraea</i>	Newly settled	1-m ² quadrats (<i>n</i> = 20) along a 40 m transect adjacent to transect used for analysis of community structure and <i>Diadema</i> populations in algal and urchin zones	10	27 months	Arnold et al. (2010)
Curaçao, Netherlands Antilles	0.1–6.7 recruits/plate	<i>Agaricia</i> , <i>Porites</i>	≤40 mm	10 × 10 × 1 cm terracotta coral settlement plates (<i>n</i> = 40) deployed inside and outside damselfish territories	3–9	Single	Bak and Engel (1979)
	16.8 recruits (m ²)	<i>Agaricia agaricites</i> , <i>Leptoseris cucullata</i> , <i>Porites astreoides</i> , <i>Meandrina meandrites</i> , <i>Agaricia lamarcki</i>	≤40 mm	1-m ² quadrats (<i>n</i> = 15/depth)	9–17		
	12.9 recruits/m ²				17–26		
	13.7 recruits/m ²				26–37		
	17.9 recruits/m ²				12 (1999)		Vermeij (2006)
	Total of 0 recruits/rack	<i>Agaricia</i> , <i>Agaricia lamarcki</i> , <i>Astrangia solitaria</i> , <i>Diploria labyrinthiformis</i> , <i>Diploria strigosa</i> , <i>Eusmilia fastigiata</i> , <i>Montastraea annularis</i> , <i>Montastraea cavernosa</i> , <i>Madracis decactis</i> , <i>Madracis pharensis</i> , <i>Meandrina meandrites</i> , <i>Stylaster roseus</i> , <i>Porites</i> , <i>Rhizosmilia maculata</i> , <i>Siderastrea siderea</i> , <i>Tubastrea coccinea</i>	Newly settled	Four formica settlement plates (each 1.2 × 0.6 m) in settlement rack (total surface area 23.12 m ²)	12 (2000)		
	Total of 33 recruits/rack				12 (2001)		
	Total of 83 recruits/rack				12 (2002)		
	Total of 278 recruits/rack				12 (2003)		
	Total of 279 recruits/rack				12 (2004)		
	Total of 251 recruits/rack				30 (1999)		
	Total of 0 recruits/rack				30 (2000)		
	Total of 89 recruits/rack				30 (2001)		
	Total of 38 recruits/rack				30 (2002)		
	Total of 127 recruits/rack				30 (2003)		
	Total of 100 recruits/rack				30 (2004)		
	Total of 107 recruits/rack				3–9	12 months	Vermeij et al. (2011)
	7.5–16 juveniles/1.25 m ²	<i>Agaricia</i> , <i>Porites astreoides</i> , <i>Siderastrea</i> , <i>Stephanocoenia intersepta</i> , <i>Madracis</i> , <i>Montastraea cavernosa</i> , <i>Agaricia lamarcki</i> , <i>Meandrina meandrites</i> , <i>Mycetophyllia</i> , <i>Scolymia</i> , <i>Eusmilia fastigiata</i> , <i>Favia fragum</i> , <i>Diploria labyrinthiformis</i> , <i>Tubastrea coccinea</i> , <i>Montastraea</i> , <i>Porites porites</i> , <i>Madracis mirabilis</i> , <i>Colpophyllia natans</i> , <i>Leptoseris cucullata</i> , <i>Dichocoenia stokesi</i>	<40 mm	0.25-m ² quadrats along a 5 m long transect (<i>n</i> = min 8/depth zone) covering a total area of 1.25 m ²	9–17		
	4–11 juveniles/1.25 m ²				17–26		
	5–12 juveniles/1.25 m ²				26–37		
Florida Keys (Key Largo)	5.5–13 juveniles/1.25 m ²				<10	Single	Chiappone and Sullivan (1996)
	1.18–3.74 colonies/m ²	<i>Agaricia agaricites</i> , <i>Porites astreoides</i> , <i>Porites porites</i> , <i>Siderastrea siderea</i> , <i>Scolymia</i> , <i>Mycetophyllia</i> , <i>Diploria clivosa</i> , <i>Montastraea cavernosa</i>	1–20 mm	1-m ² quadrats (<i>n</i> = 50/site)			
Florida Keys (Upper Keys)	1–13 recruits/0.25 m ²	<i>Siderastrea siderea</i> , <i>Porites astreoides</i> , <i>Agaricia agaricites</i> , <i>Favia fragum</i>		Permanently marked 0.25-m ² quadrats	Outer and patch reefs	12 months	Miller et al. (2000)
Florida Keys	2.75–4.5 recruits/m ²	<i>Porites astreoides</i> , <i>Favia fragum</i> , <i>Agaricia</i> , <i>Siderastrea siderea</i> , <i>Siderastrea radians</i> , <i>Diploria</i> , <i>Montastraea cavernosa</i> , <i>Porites porites</i> , <i>Montastraea faveolata</i>	≤50 mm	Field observations at restoration sites	2.5–11	Single	Miller and Barimo (2001)
Florida Keys	0–9 colonies/m ²	<i>Agariciidae</i> , <i>Poritidae</i> , <i>Faviidae</i>	Newly settled	Settling trees composed of 12 ceramic tiles (11 cm × 11 cm)	0–10	12 months	Tougas and Porter (2002)

Table 1 (continued)

Location	Density of coral recruits	Corals	Definition	Methods	Depth (m)	Duration	Reference
Florida Keys	2.1 recruits/0.25 m ²	<i>Montastraea</i> , <i>Agaricia</i> , <i>Porites</i> , <i>Siderastrea siderea</i> , others	2–40 mm	0.25-m ² quadrats placed randomly along a transect	14	Single	Edmunds et al. (2004)
	2.0 recruits/0.25 m ²				17		
	1.9 recruits/0.25 m ²				20		
	1.9 recruits/0.25 m ²				26		
Florida Keys	4–43 recruits/0.25 m ²	<i>Siderastrea radians</i>	Newly settled	Photographic monitoring of 28 permanent quadrats (50 × 50 cm)	8.5–10.7	3 months	Vermeij (2005)
Florida Keys (Key Largo)	10–38 recruits/0.25 m ²	<i>Siderastrea radians</i>	<1 mm	0.25-m ² quadrats on a shipwreck and repeated photographic censuses (3 month intervals)	Shallow fore-reef		Vermeij and Sandin (2008)
Florida Keys	Total of 2 recruits/2250 m ²	<i>Acropora palmata</i>	≤30 mm	7-m radius study plots established within an extant <i>Acropora palmata</i> stand	4–6	3 years	Williams et al. (2008)
Grenada	2–45 recruits/m ²	<i>Agaricia</i> , <i>Porites</i> , <i>Siderastrea siderea</i> , <i>Siderastrea radians</i> , <i>Stephanocoenia intersepta</i> , <i>Leptoseris cucullata</i> , <i>Madracis</i> , <i>Dichocoenia stokesi</i> , <i>Favia fragum</i> , <i>Manicina areolata</i> , <i>Diploria</i> , <i>Montastraea</i>	≤40 mm	1-m ² quadrats (<i>n</i> = 20) along a 40 m transect adjacent to transect used for analysis of community structure and <i>Diadema</i> populations in algal and urchin zones	<6	Single	Carpenter and Edmunds (2006)
Jamaica	Total of 104–339 recruits	<i>Agariciidae</i> , <i>Poritidae</i> , <i>Faviidae</i>	Newly settled	Eighty 40-cm <i>Acropora cervicornis</i> branches used as settlement substrate in each study site tied to PVC pipes in various replicated cage treatments	11–14	9–10 months	Rylaarsdam (1983)
	2–32 recruits/m ²	<i>Leptoseris cucullata</i> , <i>Agaricia agaricites</i> , <i>Tubastrea coccinea</i>	<10 mm	Twelve 1-m ² quadrats	10	12 months	Hughes (1985)
	2–12 recruits/m ²				20		
	2–45 recruits/m ²	<i>Agaricia agaricites</i> , <i>Leptoseris cucullata</i> , <i>Montastraea annularis</i> , others		1-m ² quadrats (<i>n</i> = 12)	35		Hughes and Tanner (2000)
	5–27 recruits/m ²	<i>Agaricia</i> , <i>Porites</i> , <i>Siderastrea siderea</i> , <i>Siderastrea radians</i> , <i>Stephanocoenia intersepta</i> , <i>Leptoseris cucullata</i> , <i>Madracis</i> , <i>Dichocoenia stokesi</i> , <i>Favia fragum</i> , <i>Manicina areolata</i> , <i>Diploria</i> , <i>Montastraea</i>	≤40 mm	1-m ² quadrats (<i>n</i> = 20) along a 40 m transect adjacent to transect used for analysis of community structure and <i>Diadema</i> populations in algal and urchin zones	<6	Single	Carpenter and Edmunds (2006)
St. Croix, USVI	Total of 16–38 recruits/plate	<i>Agaricia</i> , <i>Madracis decactis</i> , <i>Porites</i> , <i>Favia fragum</i> , <i>Stephanocoenia intersepta</i>	Newly settled colonies (≤31 mm)	Slabs of <i>Acropora palmata</i> (120 cm ²) attached to a plastic base secured to the substrate (<i>n</i> = 48 plates)	9	Two years	Rogers et al. (1984)
	Total of 48–54 recruits/plate				18		
	Total of 34–36 recruits/plate				27		
	Total of 13–32 recruits/plate				37		
	13–33 juveniles/m ²	<i>Agaricia agaricites</i> , <i>Porites</i> , <i>Madracis decactis</i> , <i>Agaricia lamarcki</i> , <i>Montastraea cavernosa</i>	≤40 mm	1-m ² quadrats (<i>n</i> = 20) along transect tapes	9	Single	
	28–42 juveniles/m ²				18		
	8–18 juveniles/m ²				27		
	3–5 juveniles/m ²				37		
	9–28 recruits/m ²	<i>Agaricia</i> , <i>Porites</i> , <i>Siderastrea siderea</i> , <i>Siderastrea radians</i> , <i>Stephanocoenia intersepta</i> , <i>Leptoseris cucullata</i> , <i>Madracis</i> , <i>Dichocoenia stokesi</i> , <i>Favia fragum</i> , <i>Manicina areolata</i> , <i>Diploria</i> , <i>Montastraea</i>		1 m ² quadrats (<i>n</i> = 20) along a 40 m transect adjacent to transect used for analysis of community structure and <i>Diadema</i> populations in algal and urchin zones	<6	Single	Carpenter and Edmunds (2006)
St. John, USVI	2–14 juveniles/0.25 m ²	<i>Montastraea annularis</i> , <i>Montastraea cavernosa</i> , <i>Siderastrea radians</i> , <i>Agaricia</i> , <i>Diploria strigosa</i> , <i>Porites porities</i> , <i>Porites astreoides</i>		0.25-m ² quadrats (<i>n</i> = 40) positioned randomly along a 40 m transect	5	12 months	Edmunds (2000a)
	0–20 recruits/0.25 m ²	<i>Porites</i> , <i>Agaricia</i> , <i>Favia fragum</i> , <i>Siderastrea</i> , <i>Diploria</i>	4– ≤ 40 mm	0.25-m ² quadrats (<i>n</i> = 40) visually surveyed along a 40 m transect line	≤10		Edmunds (2000b)
	3.5–5.4 recruits/0.25 m ²	<i>Siderastrea</i> , <i>Porites</i> , <i>Favia fragum</i> , <i>Agaricia</i>	2– ≤ 40 mm	0.25-m ² quadrats (<i>n</i> = 40) along a permanently marked 40 m transect	5–9		Edmunds (2004)
	0–1.5 juveniles/m ²	<i>Montastraea annularis</i>	≤40 mm	0.25-m ² quadrats censured from 1995–2004			Edmunds and Elahi (2007)
	0.1–3.6 recruits/tile	<i>Agariciidae</i> , <i>Porites</i> , <i>Faviidae</i> , <i>Siderastrea</i> , others	<2 mm	15 × 15 × 1 cm settlement tiles (<i>n</i> = 10–15/site) deployed at each location for 6 month durations (4 sampling periods)	5–6	6 months	Green and Edmunds (2011)

in the Caribbean have certainly recorded higher rates of coral recruitment than the present study: for example, Green and Edmunds (2011) reported 1.7 recruits per tile (Table 1). Recruitment studies in the Caribbean over the last three decades have generally reported a dominance of brooders, such as agariciids and poritids (Bak and Engel, 1979; Rylaarsdam, 1983; Green and Edmunds, 2011). Recruits of broadcasting coral species, such as *Acropora* and *Orbicella*, typically have been sparse (Rylaarsdam, 1983). We did not find any *Orbicella* recruits in the present study, but the 'supply' of *Acropora* recruits onto the tiles was encouraging and noteworthy.

There was a significant mismatch between the coral taxa that recruited and the adult taxa that were established on the reef (Toth et al., 2014). Although *Siderastrea*, *Porites*, and *Astrocoeniidae* were present both as adults and recruits, *Orbicella* were absent as recruits but dominant as adult colonies. By contrast, *Acropora* were common as recruits but virtually absent as adults. Recruitment of corals, therefore, does not directly equate with contribution to a given population. Previous work indicates that survival of recruits is particularly low in habitats that experience extreme environmental conditions (Bak and Engel, 1979; van Woesik et al., 1999). In 2011 the Florida Keys only supported on average 2–3% substrate coverage of living coral (Office of the National Marine Sanctuaries 2011; Toth et al., 2014). We suggest that less-than-favorable environmental conditions may cause high coral mortality after settlement, although further work is needed to identify the mechanism causing the putatively high post-settlement mortality.

In the present study, coral recruits preferentially settled on the bottoms and sides of the tiles. Only three coral recruits settled on the tops of the tiles. The preference of coral recruits for cryptic microhabitats, such as the bottom surfaces of tiles, is believed to reflect their preferences for low light levels, low sedimentation rates, and reduced levels of competition and predation (Bak and Engel, 1979; Rogers et al., 1984; Tomascik and Sander, 1987; Edmunds et al., 2004; Gleason et al., 2006). The present study also showed no relationship between the density of coral recruits and the percentage cover of macroalgae, and no relationship between recruit density and the percentage cover of coralline algae. Coral recruits did not consistently settle and metamorphose directly on the surfaces of the crustose coralline algae, but instead settled in the immediate vicinity of the coralline algae. Several other studies have also shown that coral recruits prefer to settle on the substrates adjacent to crustose coralline algae (Kitamura et al., 2007; Heyward and Negri, 1999; Ritson-Williams et al., 2010), rather than directly on the crustose corallines as suggested by earlier studies (Morse et al., 1996).

Fishing pressure had no detectable effect on the coverage of macroalgae cover on the tiles. In a 2007 study in the Florida Keys, Kramer and Heck (2007) showed that adult parrotfish were more abundant in protected areas than outside those areas, yet juvenile parrotfish were more abundant in fished areas, and the densities of surgeonfish (*Acanthuridae*) were not related to level of protection. In the Florida Keys, therefore, it seems that the small no-take reserves have increased the densities of carnivorous fishes but have neither increased nor decreased the densities of herbivorous fishes (Kramer and Heck, 2007). Equivalent densities of herbivorous fishes inside and outside no-take reserves may in part explain why we found no overall reduction in macroalgal cover on tiles placed on unfished reefs.

Similar macroalgal cover on tiles and homogeneously distributed herbivorous fishes in fished and unfished reefs do not, therefore, explain why we found more coral recruits on tiles on unfished reefs than on fished reefs. Hydrography may provide the explanation. Regularly forming gyres, particularly in the lower Florida Keys, have the capacity to concentrate larvae and may act as larval

sinks (Lee et al., 1994; Criales and Lee, 1995). It is conceivable that the recruitment patterns observed in the present study were in part a consequence of local hydrography.

In conclusion, contrary to the general assumption of limited recruitment in the Florida Keys, we found considerable densities of recruits of *Acropora* species. There was a mismatch between coral recruitment and the low coverage of coral assemblages—especially the *Acropora* species—suggesting that recruits were arriving but not surviving to contribute to coral recovery. Our results highlight the urgency of understanding the processes that are reducing the survivorship of coral recruits in the Florida Keys.

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